Module 1 Basic Content:

The natural water cycle is also known as the hydrologic cycle and the hydrogeologic cycle. Water is normally found in one of three phases: as a gas (vapor), a liquid (water), and a solid (ice). The natural water cycle involves the phase-shifting of water as it moves around planet earth – under the surface, on the surface, and above the surface of our planet – including the movement of water through plants. The water cycle includes the following:

"Bodies" of water

- Atmosphere: *gas* – water vapor (cannot see) and clouds
- Surface/above ground: *liquid* – oceans, seas, lakes, springs, swamps, bogs, ponds, rivers, streams, and all plants; *solid* – snowpack, glaciers, ice sheets
- Underground: *liquid* – ground water, aquifers

Water phases

- Gas (AKA water vapor): water molecules have enough energy from the sun that they are far apart, do not cling together, and cannot be easily seen or touched (we inhale and exhale water vapor all the time and do not notice it)
- Liquid (AKA water): water molecules cling together as droplets which can be seen and touched but have enough energy from the sun that they do not solidify; as the droplets collect they form a fluid that takes the shape of any container it is placed in
- Solid (AKA ice or snow): water molecules have lost enough energy that they harden into a solid that can be seen and touched, and holds its shape without a container

Water phase-*shifts*

- Evaporation: *phase-shift* from liquid to gas as water molecules leave a body of water and move into the atmosphere in the form of vapor; energy from the sun heats the molecules so they move fast enough to make the phase-change and escape as a vapor from the water's surface
- Transpiration (AKA evapotranspiration): *phase-shift* from liquid to gas as water molecules leave the stomata on the underside of plant leaves and move into the atmosphere in the form of water vapor; energy from the sun is required to heat the molecules so they move fast enough to make the phase-change

    - Odd factoid: Transpiration is simply evaporation from plants. The reason it has a name is because there are so many plants on planet earth that transpiration makes up a significant portion of the evaporative process and we need to be aware of that fact.
    - Odd factoid: Transpiration must occur so water will move up through the roots of plants – as water escapes from the leaves it draws up water from the surrounding soil through the roots (similar to sipping a straw). This is why plants wilt when the soil is dry, because water has escaped from the leaves but has not been replaced from the roots, causing all the cells to shrink due to lack of water and the plant to droop as a result.
- Condensation: *phase-shift* from gas to liquid as water molecules in vapor release energy to the atmosphere and cool off so they stick together and form droplets - which make up clouds (and the steam we see on the mirror when we take a hot shower)
- Freezing: *phase-shift* from liquid to solid as water molecules release enough energy to slow down their movement so the molecules stick together in a hard matrix called ice
    - Odd factoid: there are many forms of frozen precipitation and all of them are composed of ice in various formats from pellets of ice called sleet to snow and frost which are both made of ice crystals connected in dainty patterns
    - Odd factoid: ice is less dense than water so a given amount of water fills less space than it will when frozen – this is why ice floats in water, why frost wedging occurs, and why water pipes break when the water in them freezes
- Sublimation: *phase-shift* from solid to gas, without going through the liquid phase; water molecules are removed from ice and released as water vapor, important: rather than the sun's energy fueling this phase-shift it is usually fueled through wind action
        - Odd factoid: sublimation is the reason why the ice cubes in our freezers shrink in the tray and will eventually disappear if left alone for a few weeks – the ice cubes have not melted but the molecules have simply been "blown away" as the freezer fan kicks on and off
        - Odd factoid: sublimation is also the reason why the ice on roadways sometimes "disappears" overnight when the temperatures are too low

for the ice to melt but it is breezy – the wind sublimated the ice on the roadway by picking up frozen water molecules from the surface of the ice and carrying them away

Water cycle processes

- Transport processes: the cycled movement of water molecules from one place to another on the earth, occurring in the atmosphere, on the surface, and below the earth's surface; above ground transport is fueled by the sun and gravity, below ground transport is fueled by gravity and differences in pressure.

All of the following are different forms of transport processes.

- Evaporation: water molecules in bodies of water move from the surface of the water into the atmosphere as a vapor
    - Odd factoid: evaporation and is one of the ways in which water is purified of impurities since individual water molecules do not carry other molecules with them when they leave the water surface as vapor
- Transpiration: water molecules in plants (xylem) move from the leaf (stomata) of the plant into the atmosphere as a vapor
- Condensation: water molecules in the form of vapor in the atmosphere stick together and form clouds made of water droplets
- Precipitation: water droplets in clouds combine and eventually become heavy enough to fall from the atmosphere to the earth's surface; precipitation takes many forms like rain, mist, snow, sleet, hail, etc., so sometimes the water droplets phase-shift from liquid to solid (ice or snow) before falling to the earth's surface
- Run-off: precipitation lands on the surface of the earth and flows from higher elevations to lower elevations due to gravity; run-off is what typically fills surface bodies of water
- Infiltration: precipitation lands on the surface of the earth and soaks into the soil; this water may be taken up by plants through their roots or may move further down and percolate into an aquifer
    - Odd factoid: infiltration is one of the ways in which water is purified of impurities through both filtration (sieves out larger particles), and adsorption (as water moves through the soil, impurities dissolved in the water stick to the particles soil, removing the impurities from the water);

it seems like a strange way to clean water since we tend to think of soil as “dirty”, but this is also how carbon filters work in fish tanks

- Percolation: movement of water droplets deeper into the soil and rock where it becomes groundwater; groundwater may flow freely through the soil and rock or be confined in an aquifer – the top of this “flow” of groundwater is called the water table
    - Odd factoid: percolation, like infiltration, also cleans water through filtration and adsorption
    - Odd factoid: springs and seeps are points where the water table breaks the surface and allows groundwater to flow out of the ground; this water often feeds creeks or ponds, providing a source of water even in times of drought
    - Odd factoid: artesian springs are different in that they are fed by a hole in a confined aquifer and are under such pressure that the water is forced up through the rock to the surface; a few artesian wells are under such pressure that rather than simply bubbling to the surface they shoot up into the air